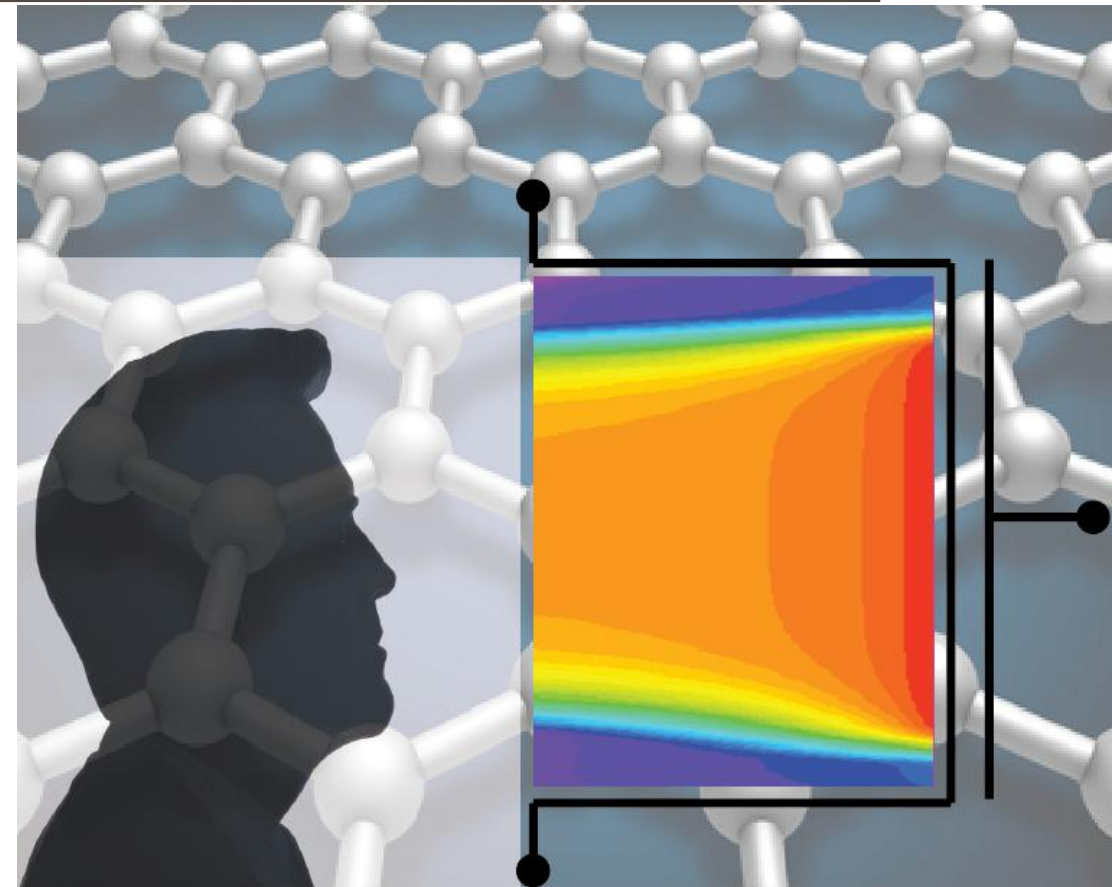


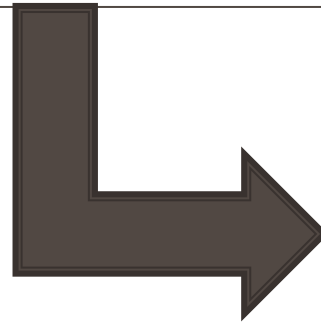
ADVANCED SOLID-STATE DEVICES

Week-13, Lecture-1

Sneh Saurabh
17th April, 2025



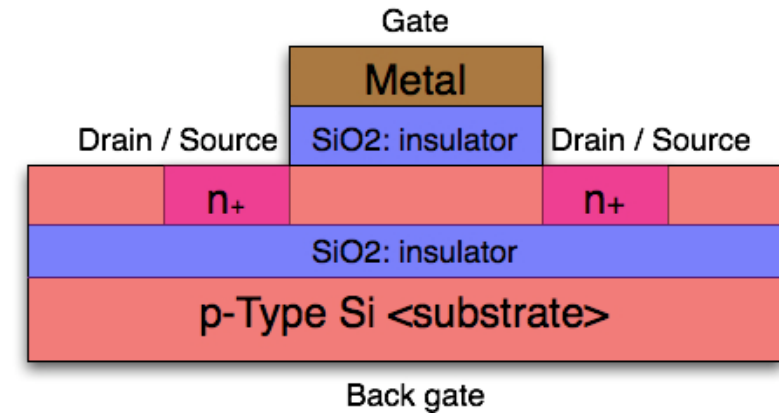
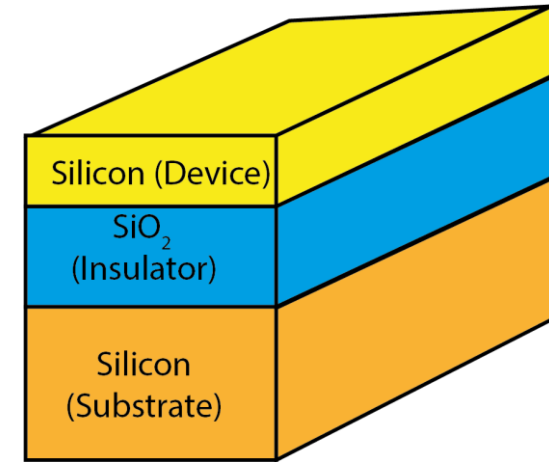
Advanced MOSFETs



FD- SOI
MOSFETs

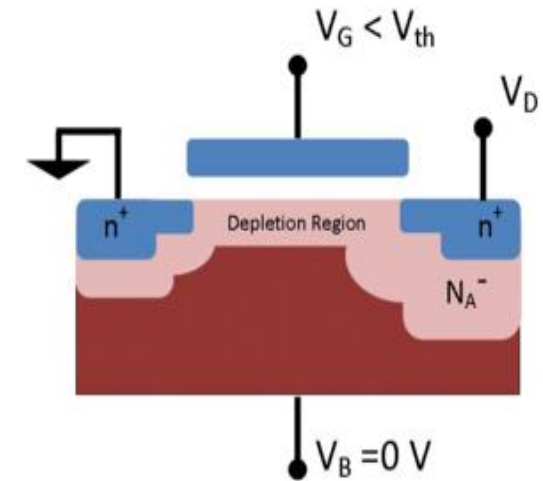
Silicon on Insulator

- FD-SOI FET: **Fully Depleted Silicon on Insulator** Field-Effect Transistor
- SOI: **Silicon on Insulator**
- SOI consists of a top thin layer of silicon and a bottom thick layer of silicon substrate separated by an insulator material such as silicon dioxide (SiO_2)
- Devices are built on the top silicon layer
- SOI wafers are categorized into two types:
 - **Thick-Film SOI**: Partially Depleted SOI
 - **Thin-Film SOI**: Fully-Depleted SOI

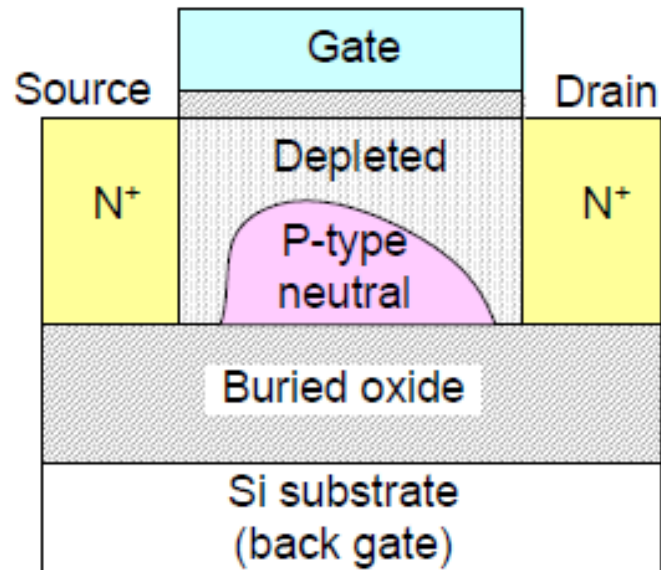


Depletion layer in MOSFET

- Depletion layer is formed at the boundary of PN junction
- Depletion region formed below the gate under equilibrium
- The width of the depletion region changes with gate bias
- For p-type substrate:
 - As +ve voltage is applied to the gate, holes in P-substrate are repelled away from the gate, leaving behind region that are depleted of carriers
 - As gate voltage is increased depletion region width goes on increasing and *attains a maximum*

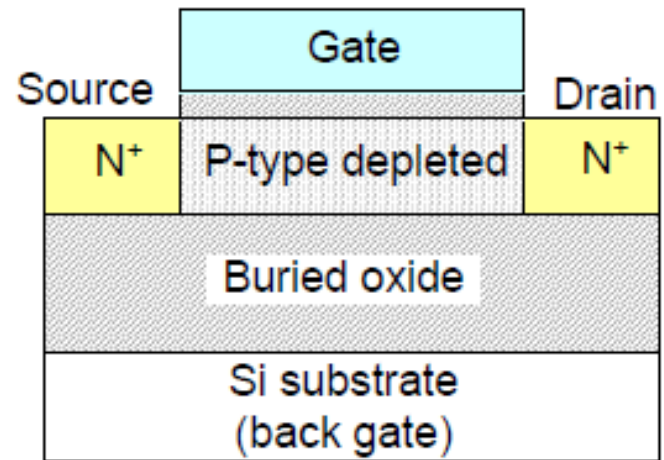


Partially Depleted SOI



- In SOI both top-gate and substrate [back gate] controls the potential in the channel
 - There are two depletion regions: below each gates
-
- **Thick-film SOI**
 - Silicon film thickness is larger than $2 \times x_{d-max}$
 - Two depletion regions arising due to front and back interfaces do not interact
 - There lies a neutral region between two depletion regions
 - This is called Partially Depleted SOI
 - If the neutral region is “somehow” connected to ground, the behavior will be same as a traditional bulk MOSFET

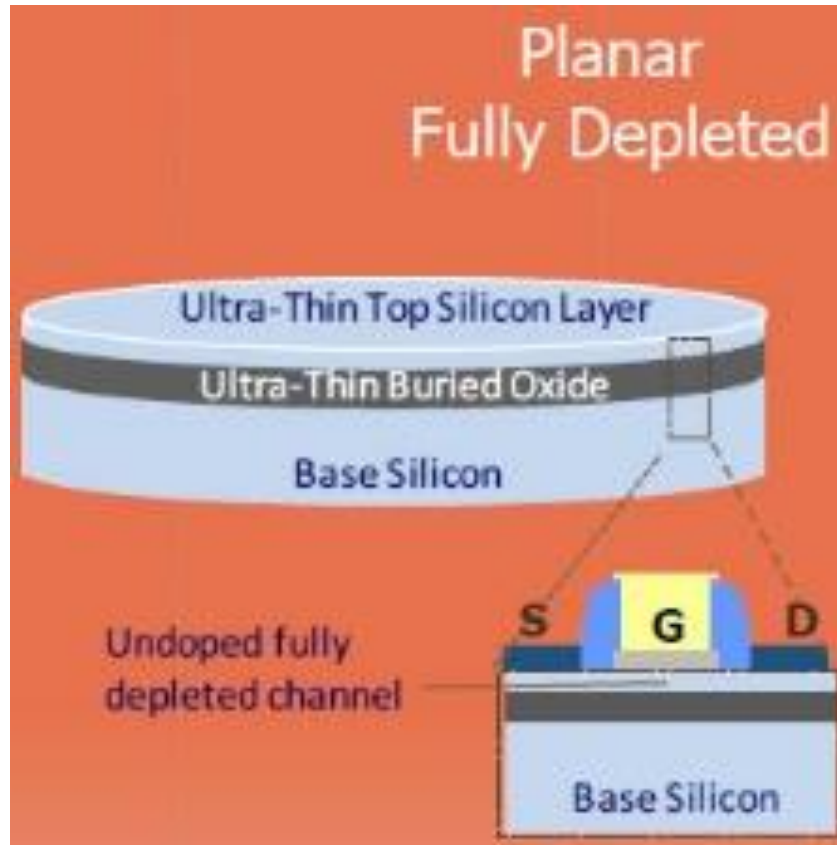
Fully Depleted SOI (FD-SOI)



▪ Thin-film SOI

- Silicon film thickness is smaller than x_{d-max}
- Two depletion regions arising due to two interfaces interact and complete silicon body is fully depleted
- This is called Fully Depleted SOI [FD-SOI]
- FD-SOI exhibit better short-channel behavior and subthreshold swing

FD-SOI FET



Source:

http://www.eetimes.com/author.asp?section_id=36&doc_id=1286295

- FD SOI FET realized on an ultra-thin layer of silicon over a Buried Oxide layer

Typical Dimensions:

- Buried oxide layer: 25nm thick
- Silicon Film Thickness: 5nm-10nm silicon film with very tight tolerances [5 Angstroms across the wafer]
- Tight tolerances creates less variability in the production process leading to higher yields
- No doping is required in the body of FD-SOI

FD-SOI FET vs. Bulk MOSFET: Short Channel Effects



Source: http://www.st.com/content/st_com/en/about/innovation---technology/FD-SOI/learn-more-about-fd-soi.html

- Thin layer of silicon in FD-SOI FET allows better electrostatic control of the gate compared to bulk MOSFET
 - **Reason:** The part of device that is away from the gate is no more active in FD-SOI FET
 - **Effects:**
 - ✓ Better short-channel characteristics
 - ✓ Sharper subthreshold swing
 - ✓ Lower OFF-state current

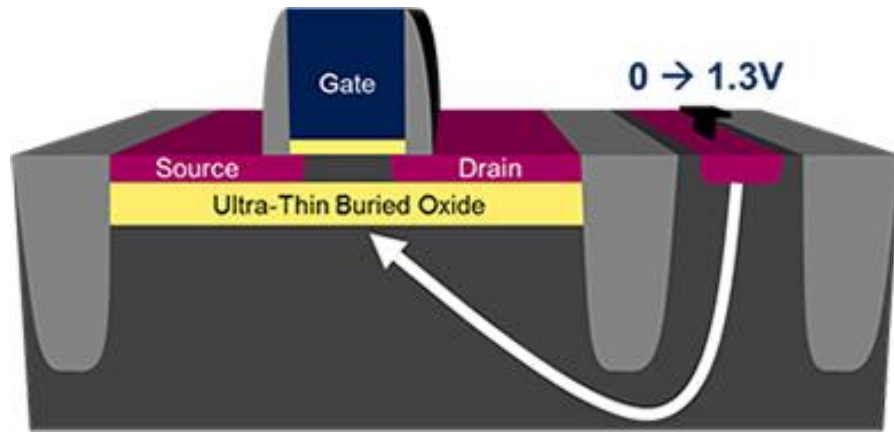
FD-SOI FET vs. Bulk MOSFET: Capacitances



Source: http://www.st.com/content/st_com/en/about/innovation---technology/FD-SOI/learn-more-about-fd-soi.html

- Parasitic capacitances such as junction capacitance between drain (source) to substrate (well) is reduced in FD-SOI FET
 - The *speed* of circuits realized using FD-SOI FET is high

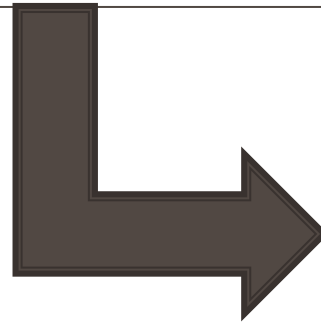
FD-SOI FET: Substrate Control



Source:
http://www.st.com/content/st_com/en/about/innovation---technology/FD-SOI/learn-more-about-fd-soi.html

- Biasing is efficient in FD-SOI FET due to thin buried oxide
- Control of substrate over the channel can be considered as a control of “back-gate”
- Presence of the buried oxide allows the application of higher biasing voltages
- *Dynamic control of the threshold voltage* of the transistor can be achieved using substrate

Advanced MOSFETs

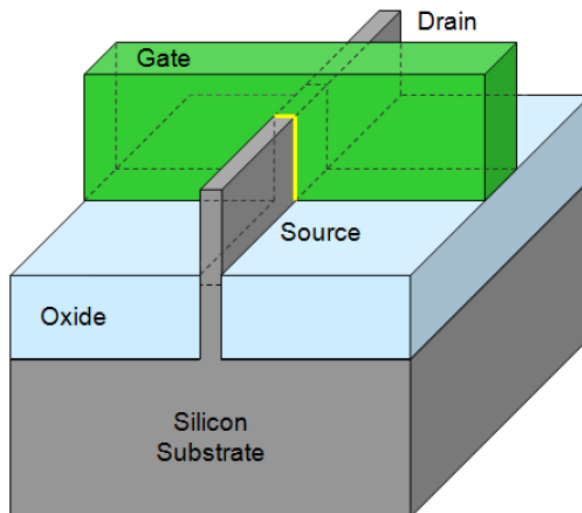


FinFETs

FinFETs: Basics



Source: Global foundries http://www.eetimes.com/document.asp?doc_id=1262552

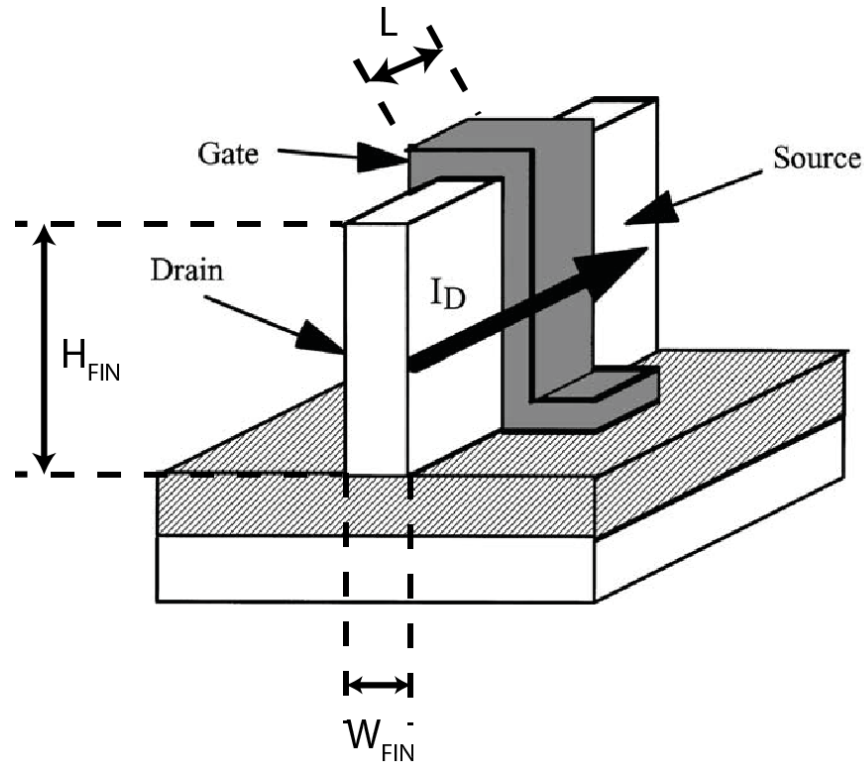


Source: Intel
<http://www.intel.com/content/www/us/en/silicon-innovations/standards-22nm-3d-tri-gate-transistors-presentation.html>

- FinFETs are *non-planar transistor* and structurally quite different from planar MOSFETs
- Term FinFET was coined by **Chenming Hu, Tsu-Jae King-Liu and Jeffrey Bokor** in 2001

- Intel calls FinFETs as tri-gate transistor and have used it in **22 nm technology** and below
- Global Foundries, TSMC are using FinFET in **14 nm/16 nm technology** nodes
- FinFETs, Double-Gate, Tri-Gate Transistors are examples of Multi-gate transistors

FinFETs: Structure



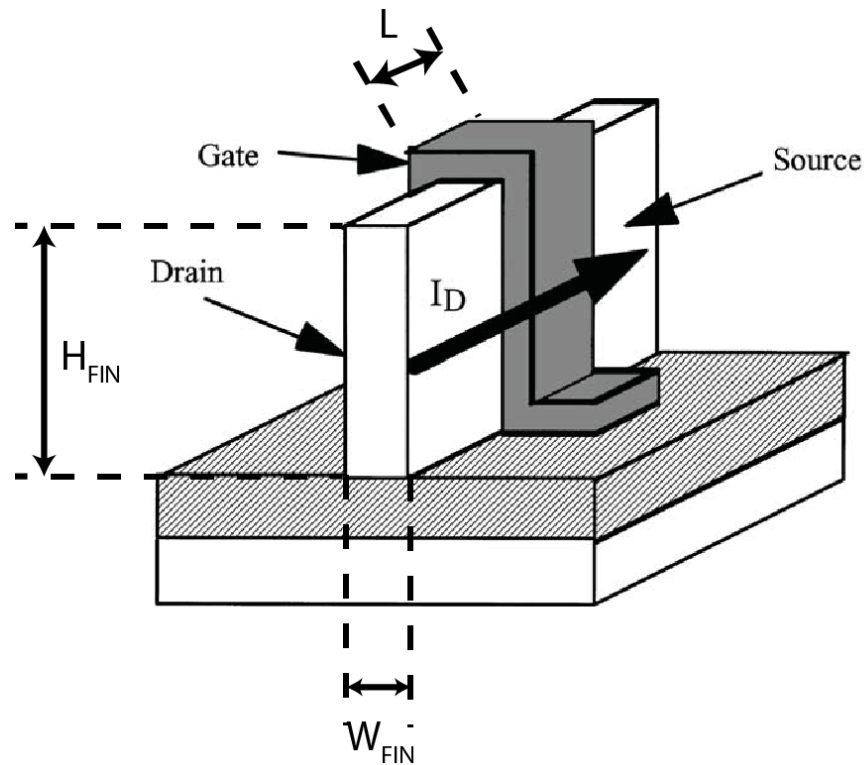
- FinFET is a non-planer or 3D transistor
- FinFET can be fabricated on bulk substrate or SOI substrate
- *Fin*: Thin silicon body protruding from the substrate forming the channel

- Tri-gate: *One gate on top* of the fin and *Two gates on the sides* of the fin

- Top-Gate: $Length = L$ $Width = W_{FIN}$
- 2 Side-Gates: $Length = L$ $Width = H_{FIN}$
- Drive Current $\propto (W_{FIN} + 2 \times H_{FIN})$

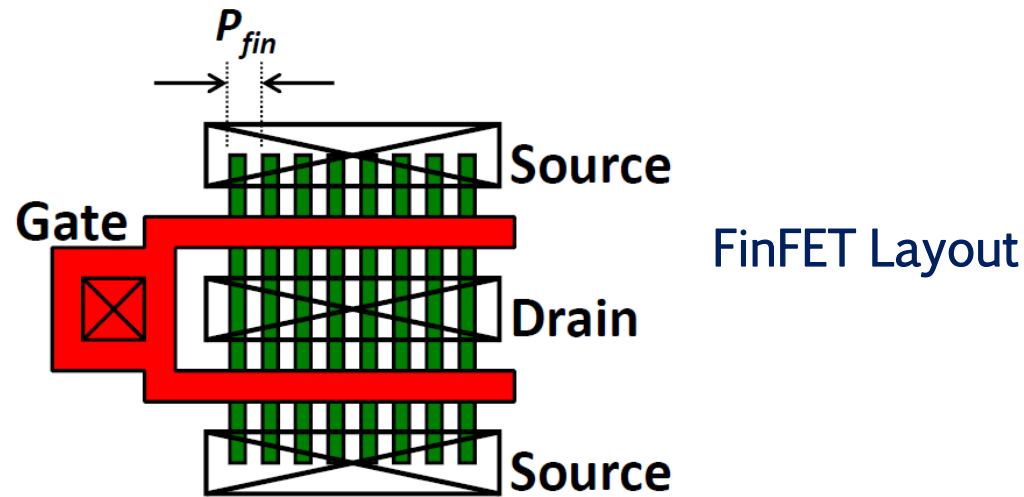
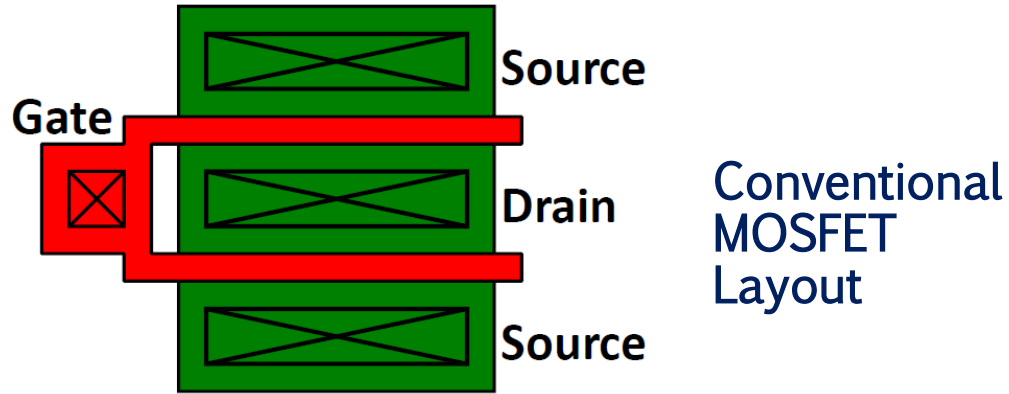
- Double-gate: If there is no gate on the top of the fin, then the structure reduces to Double-gate FET [DGFET]

FinFETs: Short Channel Effects



- Short Channel performance of Tri-Gate or FinFET is related to $\frac{L}{W_{FIN}}$
 - To adequately suppress DIBL: $\frac{L}{W_{FIN}} > 1.5$
 - FinFET will be more tolerant to short channel effects if fin is **narrow**
 - FinFET will be more tolerant to short channel effects if channel length is **longer** [obvious!]
- The scaling of W_{FIN} provides an additional **knob to improve gate control of transistor** with scaling of channel length [in addition to gate oxide scaling]

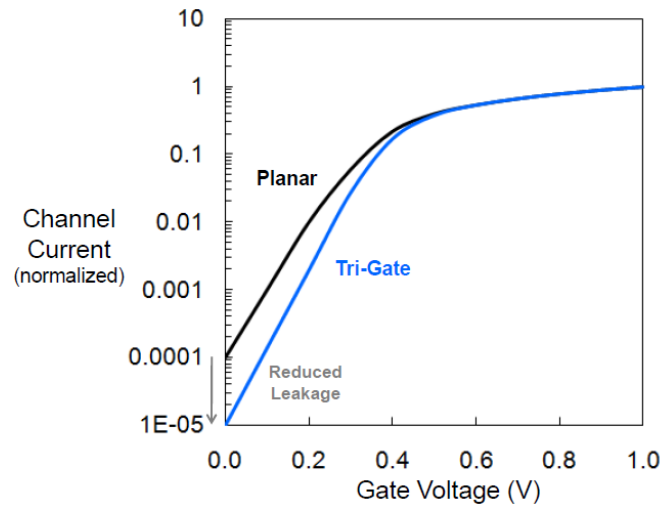
FinFETs: Layout



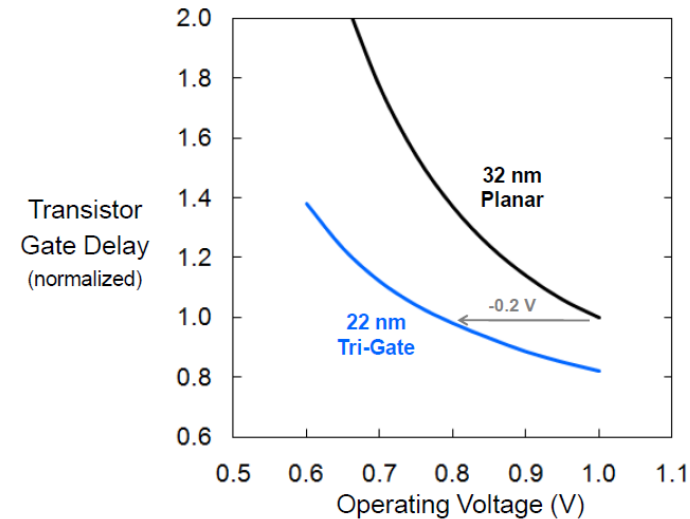
- Layout of FinFET based devices and layout of conventional MOSFETs are similar
- However width in a conventional MOSFET can be continuous and in a FinFET based layout width can have only quantized value
 - Designer can choose only integral number of fins in FinFET layout
- Separation between two fins in FinFET based layout is known as the *pitch of the FinFET* [P_{fin}]

Source: Liu, Tsu-Jae King. "FinFET history, fundamentals and future." *Proceedings of the Symposium on VLSI Technology Short Course*. University of California, 2012

FinFETs: Electrical Characteristics



- Better electrostatic control of gate
 - Sharper (Lower) subthreshold swing
 - Lower OFF-state current [Leakage Current]
 - Lower short-channel effects



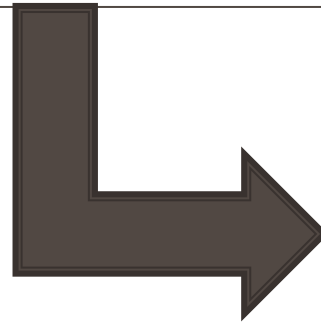
- FinFETs show *smaller delay at high voltage* and dramatically *lower delay at smaller voltage* compared to planar MOSFET
- FinFETs can operate at lower voltage for same performance [delay] => smaller dynamic power in FinFETs

Source: Intel [<http://www.intel.com/content/www/us/en/silicon-innovations/standards-22nm-3d-tri-gate-transistors-presentation.html>]

FinFETs vs. FD-SOI FETs

	FinFET	FD-SOI FET
Cost of expensive thin-film SOI wafer	Not required	Required
Fabrication Cost	New steps required in fabrication increases fabrication cost	Can reuse existing fabrication technology and hence fabrication cost is low
Ease of migration	Migrating to FinFET requires more capital	Migrating to FD-SOI FET requires less capital
Design effort	More	Similar
Drive Current	Can be higher	Limited by ultra-thin body of FD-SOI FET

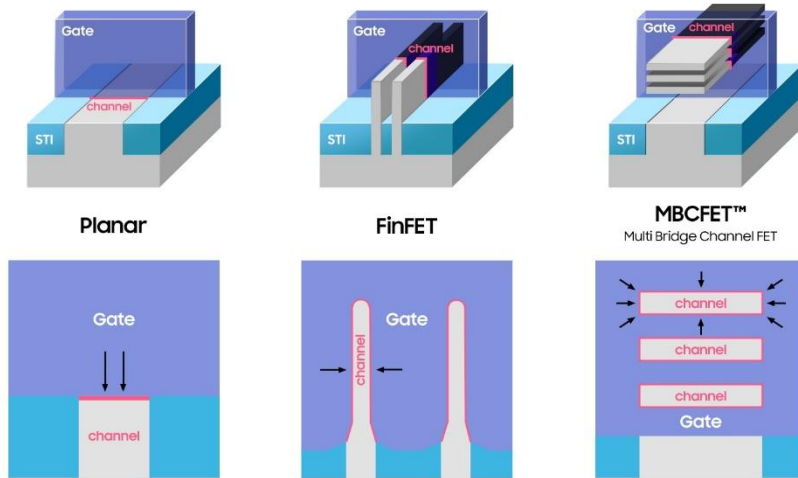
Advanced MOSFETs



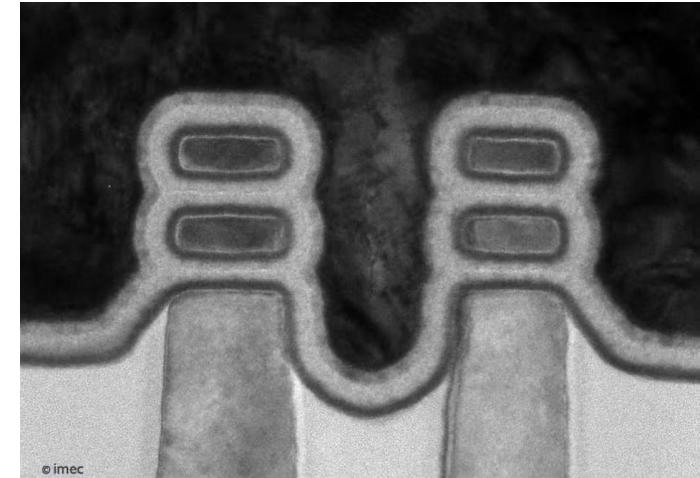
Nanosheet FETs
and CFETs

Nanosheet FETs

More Moore – Gate All Around



Source:
<https://semiconductor.samsung.com/news-events/tech-blog/3nm-gaa-mbcfet-unrivalled-sram-design-flexibility/>



Nanosheet FETs

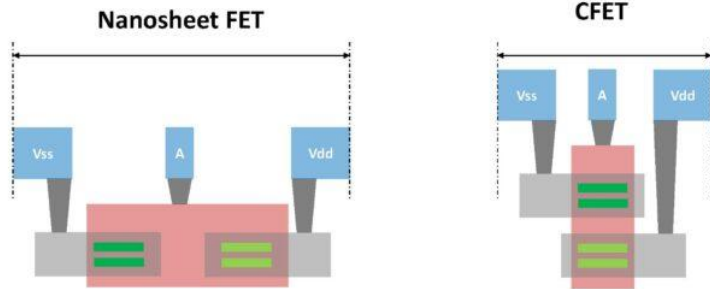
- Gate wraps around thin sheets (3-5 nm) of semiconductor stacked vertically
- Gate-all-around (GAA) structure: less short channel effects
- Larger drive current for smaller footprint: current proportional to the total width of sheets

In technologies nodes below 5 nm

- Samsung: 3nm (MBCFET: Multi-bridge Channel FET) – already manufactured in 2022
- TSMC: expected in 2 nm node (N2)
- Intel: expected in 20A (RibbonFET)

CFETs

Complementary FET (CFET)



CFET has **stacked NMOS-PMOS** architecture minimizing CMOS device foot-print

Source: <https://semiwiki.com/semiconductor-services/340811-iedm-2023-imec-cfet/>

CFETs

- CFETs: Complementary FETs
- n-type and p-type devices are vertically stacked on top of each other

Advantages:

- Reduces footprint (no need for side-by-side layout)
- Shorter local interconnects (performance improvement)

Challenges:

- Fabrication difficulties
- Parasitic (coupling) capacitances and resistance
- Complex routing
- In technologies nodes below 2 nm (around 2032)
- IMEC, TSMC, and Samsung working on it

Scaling: FinFET vs. Nanosheet FET vs. CFET

How to increase the density of transistors?

1. Pitch Scaling
2. Track Scaling

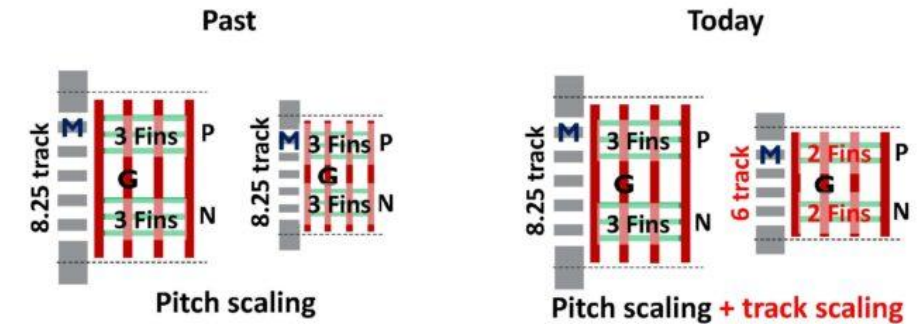
Pitch Scaling

- Reducing distance between nearby features (such as gates of transistors or metal lines)

Track Scaling

- Track is a vertical routing channel (the width of a metal line plus spacing)
- Reducing the height of standard cells by using fewer tracks
- Can lead to performance degradation

CMOS standard scaling trend



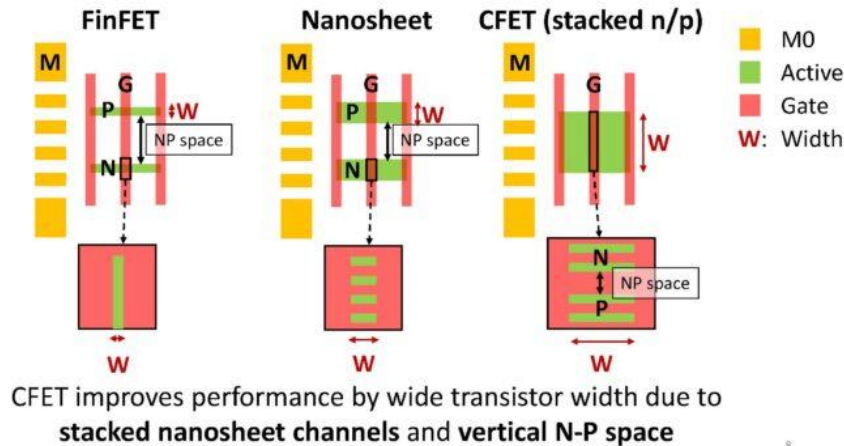
CMOS standard cell scaling today driven by pitch scaling + track scaling
(metal #/standard cell)

→ Performance degradation risk by Fin # reduction

Source: <https://semiwiki.com/semiconductor-services/340811-iedm-2023-imec-cfet/>

Scaling: FinFET vs. Nanosheet FET vs. CFET

CFET improves performance in track scaling

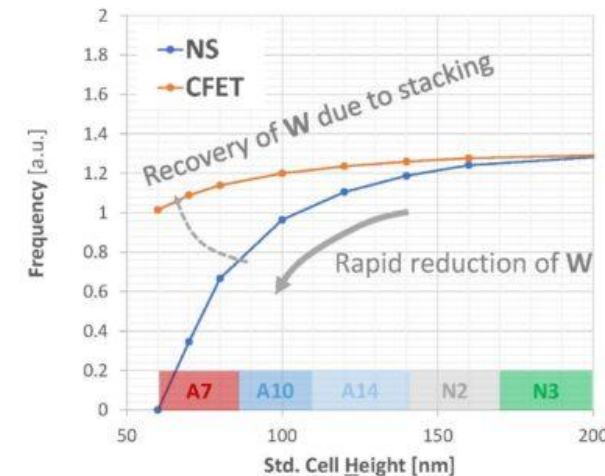


Source: <https://semiwiki.com/semiconductor-services/340811-iedm-2023-imec-cfet/>

How Nanosheet FETs help in increasing transistor density?

- Wider nanosheets are allowed because multiple nanosheets can be stacked vertically (in contrast to narrower Fins)

Performance scaling: NS vs CFET



How CFETs help in increasing transistor density?

- CFETs avoid lateral scaling constraints because the n-p devices are vertically stacked
- Enables wider sheets (recovers lost performance of scaled nanosheets)